Technical Report: Morphological Analysis of Palmer Archipelago Penguins

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Abstract

This report evaluates the morphological variations among three penguin species (*Adelie*, *Chinstrap*, and *Gentoo*) using the Palmer Penguins dataset. Through Python-based data cleaning and exploratory analysis, a strong correlation between flipper length and body mass was identified. The results demonstrate that *Gentoo* penguins are significantly larger, with an average mass of 5,067.7g, compared to the 3,703.0g and 3,733.1g observed in *Adelie* and *Chinstrap* species, respectively. These findings suggest that flipper length and body mass are reliable metrics for identifying *Pygoscelis papua* in field observations.

Keywords: Data Cleaning, Exploratory Data Analysis (EDA), Palmer Archipelago, *Pygoscelis*, Morphological Variation, Python Programming, Statistical Correlation, Biological Classification

1 Introduction

The study of morphological variation is essential for understanding how different species within the same genus adapt to their environment. In the Palmer Archipelago, Antarctica, three species of penguins—*Adelie*, *Chinstrap*, and *Gentoo*—share overlapping habitats but exhibit distinct physical characteristics [1]. This report utilizes Python-driven data analysis to explore these physical differences, focusing on the relationship between body mass and skeletal dimensions as documented in recent ecological studies [2].

2 Objectives

The primary goal of this analysis is to identify key physical differentiations between penguin species in the Palmer Archipelago. Specifically, this study aims to:

1. Perform automated data cleaning by implementing a Python script to identify and handle missing values in a raw biological dataset.
2. Analyze statistical correlations to determine the strength of the relationship between flipper length and body mass across the population.
3. Evaluate Species Classification by quantifying species-specific morphological differences to

determine if physical measurements can accurately distinguish the *Gentoo* species from its counterparts.

2.1 Hypothesis

It is hypothesized that *Gentoo* penguins will exhibit a distinct morphological profile—characterized by significantly higher body mass and longer flipper length—compared to the more similarly sized *Adelie* and *Chinstrap* species.

3 Methods

3.1 Data Source

The data used in this study is the Palmer Penguins dataset, originally collected by Dr. Kristen Gorman and the Palmer Station, Antarctica LTER [1]. It was retrieved for this analysis via the official GitHub repository provided by the dataset curators [2, 3].

3.2 Preliminary Inspection (Uncleaned Data)

Upon initial loading, the raw data (`penguins_raw.csv`) is significantly more complex and unrefined than the final version used in this report. An inspection revealed several integrity issues, including critical gaps in the form of missing values for body mass, flipper length, and sex. The raw dataset contains 17 original columns, including technical metadata such as study names, sample numbers, and individual IDs, which required immediate pre-processing to ensure the validity of the study.

3.3 Data Cleaning Process

The cleaning process involved a multi-step data pipeline executed using the Python Pandas library to transform the raw field notes into a usable technical artifact. Feature selection was prioritized, reducing the 17 original columns down to 8 core biological variables by removing metadata that did not assist in species differentiation, such as egg dates and comments. Numerical missing values were handled through median imputation to preserve the sample size without introducing outlier bias, while categorical gaps in the sex column were standardized as 'Unknown.' Furthermore, the raw taxonomic names were simplified (e.g., from *Pygoscelis adeliae* to *Adelie*) for clearer labeling.

Upon completion, the refined dataset was exported to `cleaned_penguin_data.csv` to ensure a reproducible foundation for analysis.

4 Results

The analysis of the cleaned data revealed clear morphological distinctions, primarily driven by the *Gentoo* species.

Table 1: Summary Statistics of Body Mass (g)

Species	Count	Mean Mass (g)	Max Mass (g)
Adelie	152	3,703.0	4,775.0
Chinstrap	68	3,733.1	4,800.0
Gentoo	124	5,067.7	6,300.0

4.1 Species Distribution and Class Balance

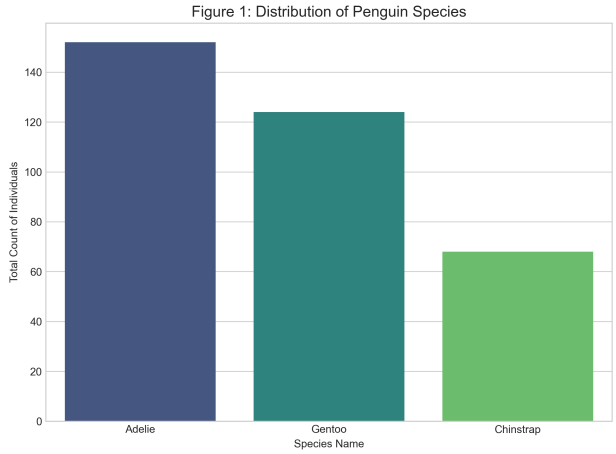
The distribution of the sample population, as illustrated in Figure 1, serves to visualize the ”class balance” of the dataset. The *Adelie* species holds the highest representation with 152 individuals, which is nearly double that of the *Chinstrap* species (68 observations). This indicates that the morphological findings for the *Adelie* population are supported by a significantly larger sample size within this specific dataset.

4.2 Morphological Correlation and Clustering Analysis

The relationship between physical dimensions was further analyzed through a correlation study of body mass and skeletal length. As demonstrated in Figure 2, a clear positive linear correlation exists between flipper length and body mass; as the data points migrate from the bottom-left to the top-right of the plot, it confirms that longer flipper dimensions generally correspond to heavier body mass across the genera.

Furthermore, Figure 2 illustrates a distinct ”clustering” effect based on species. The *Gentoo* penguins (represented by orange markers) are grouped almost entirely in isolation on the right side of the graph, visually proving the hypothesis that this species is morphologically distinct. Conversely, the markers for *Adelie* (blue) and *Chinstrap* (green) species are heavily intermixed. This overlap confirms that mass and flipper length alone are insufficient for distinguishing these

two species, necessitating the inclusion of secondary metrics, such as bill depth, for accurate classification.



5 Discussion

The research scope focused on three primary areas: morphological differentiation, size correlation, and the distinct ecological niche of the *Gentoo* species. While the raw data contained complex chemical isotope measurements, this study narrowed its focus to physical size variations to prove that flipper length increases proportionately with body mass in a predictable linear fashion.

The results strongly support the hypothesis and align with established ecological dimorphism observed in the genus *Pygoscelis* [1]. Specifically, the data highlights the ”Gentoo Niche”; even though *Adelie* and *Chinstrap* penguins are similar in size, *Gentoo* penguins occupy a much larger size niche, averaging roughly 1.3kg heavier. This transition from a broad set of field observations to a targeted comparison confirms that skeletal and mass metrics are highly effective for identifying *Pygoscelis papua*.

5.1 Limitations

While *Gentoo* penguins are easily identified by these metrics, the significant overlap in body mass and flipper length between *Adelie* and *Chinstrap* species suggests that these two cannot be distinguished by these two metrics alone in a field setting.

5.2 Conclusion

The findings of this study successfully support the initial hypothesis, as the *Gentoo* species exhibited a distinct morphological profile characterized by significantly higher body mass and longer flipper length compared to the *Adelie* and *Chinstrap* species. Flipper length and body mass proved to be highly effective predictors for identifying *Gentoo* penguins (*Pygoscelis papua*), as this group occupied a non-overlapping size niche in the correlation analysis. However, while the hypothesis was confirmed for the *Gentoo* population, the significant overlap between *Adelie* and *Chinstrap* penguins indicates that these two species cannot be distinguished by mass and flipper length alone. Consequently, for a more robust classification model covering all three species, future analysis should incorporate additional variables such as bill depth or genetic markers to improve differentiation between the more similarly sized genera.

References

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[2] A. M. Horst, A. P. Hill, and K. B. Gorman. 2020. *palmerpenguins: Palmer Archipelago (Antarctica) penguin data*. R package version 0.1.0. `https://allisonhorst.github.io/palmerpenguins/`

[3] Palmer Station, Antarctica LTER. *Palmer Archipelago (Antarctica) Penguin Data*. Retrieved March 2026 from `https://raw.githubusercontent.com/allisonhorst/palmerpenguins/master/inst/extdata/penguins.csv`